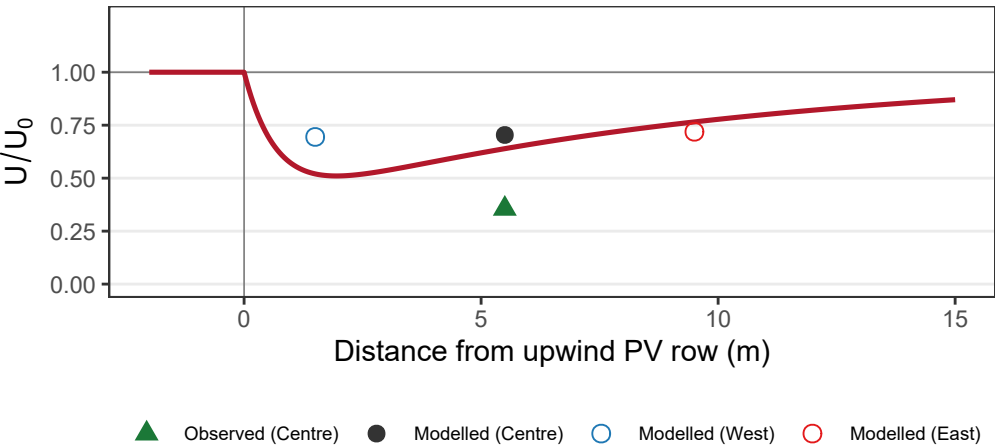


Barrier-profile mechanism (a-c)

a

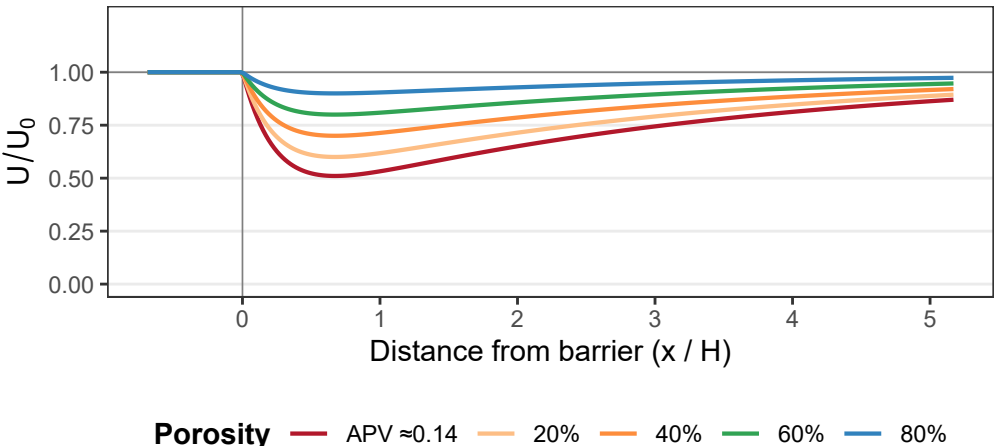
Barrier-profile reduction ($\beta_{APV} \approx 0.14$)



Points: mean U/U0 for perpendicular/oblique wind (E, W, NE, SE, NW, SW); parallel N/S excluded (Ru=1 by construction).

b

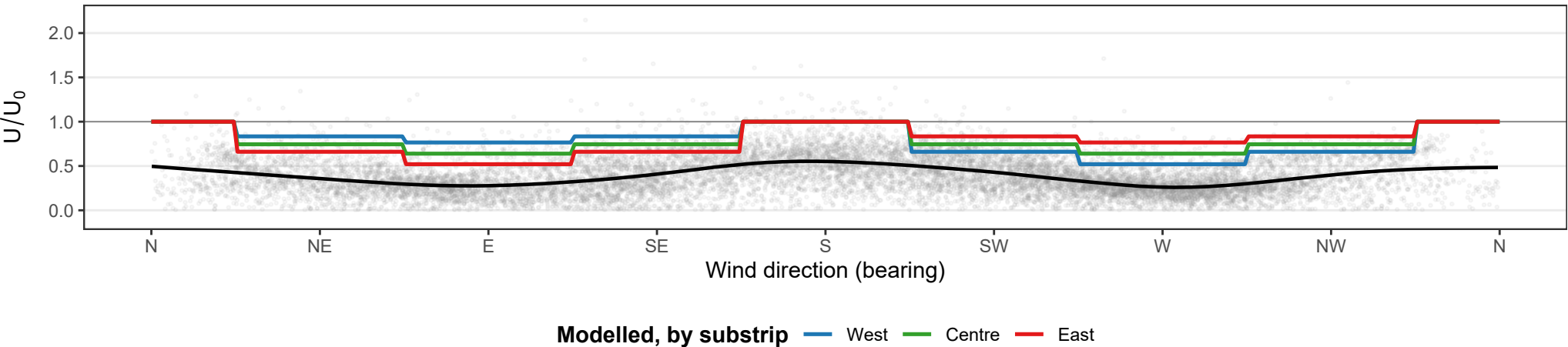
Sensitivity to panel porosity



c

Direction-aware sheltering by substrip position

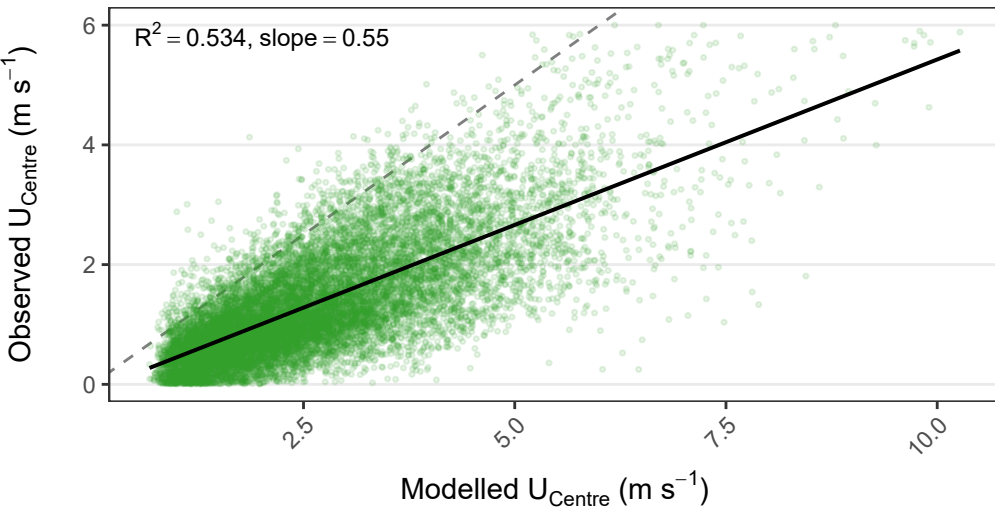
Grey points/black smooth: observed Centre. Coloured lines: modelled, all three substrips.



Empirical wind validation. Centre only (d-f)

d

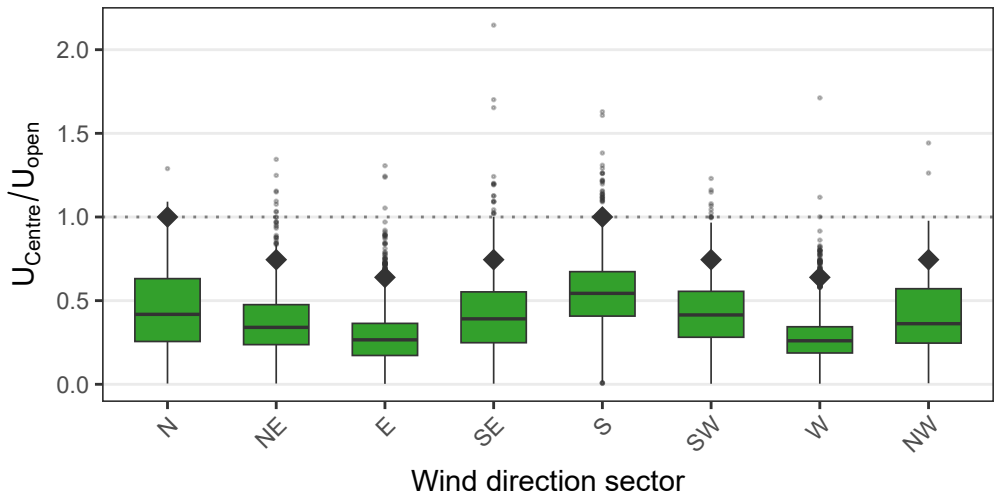
Modelled vs observed Centre wind speed



e

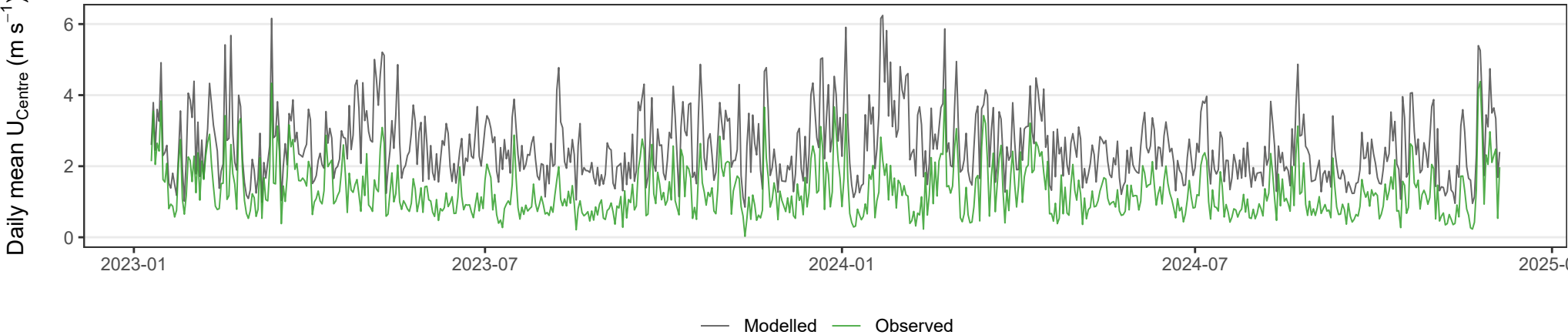
Agreement by wind-direction sector

Boxplots: observed. Diamonds: modelled.



f

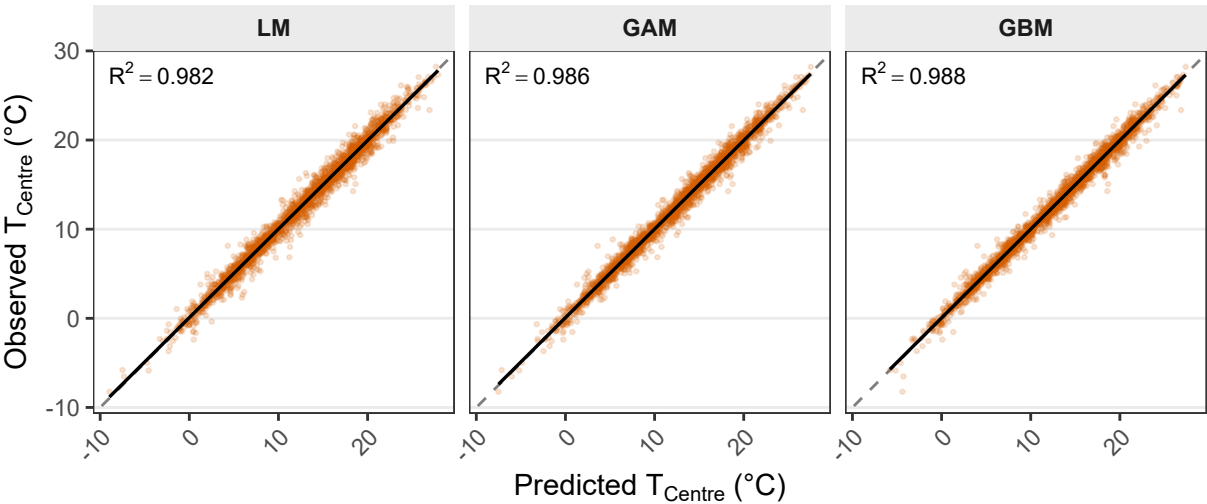
Modelled vs observed Centre wind speed over time



Temperature model comparison (g-i)

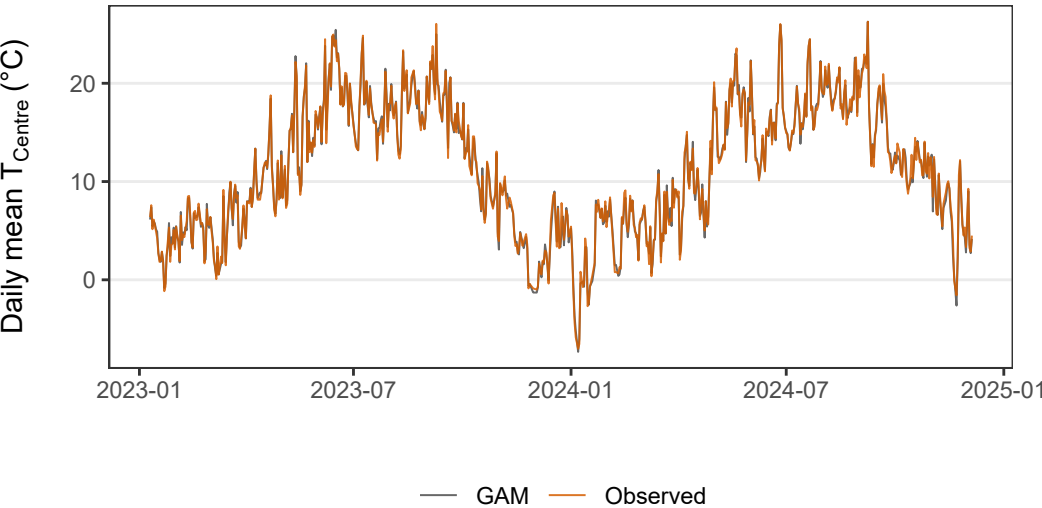
g

Held-out validation: LM vs GAM vs GBM



h

GAM prediction vs observed (test split)



i

Monthly climatology: GAM vs observed

